(ii) find the conditional probability A_3 given G that is, $P(A_3 / G)$

We have

$$P(A_1) = 0.50, \quad P(G / A_1) = 0.90$$

$$P(A_2) = 0.30, \quad P(G / A_2) = 0.70$$

$$P(A_3) = 0.20, \quad P(G / A_3) = 0.60.$$

(i) Since A_1, A_2 and A_3 are mutually exclusive and exhaustive events and G is an event in S, then the total probability of event G is P(G).

$$P(G) = P(A_1)\, P(G / A_1) + P(A_2)\, P(G / A_2) + P(A_3)\, P(G / A_3)$$

$$P(G) = (0.50)(0.90) + (0.30)(0.70) + (0.20)\ (0.60)$$

$$P(G) = 0.78.$$

(ii) The conditional probability A_3 given G is $P(A_3 / G)$

By Bayes' theorem,

$$P(A_3 / G) = \frac{P(A_3)\, P(G / A_3)}{P(A_1)\, P(G / A_1) + P(A_2)\, P(G / A_2) + P(A_3)\, P(G / A_3)}$$

$$P(A_3 / G) = \frac{(0.20)(0.60)}{(0.50)(0.90) + (0.30)(0.70) + (0.20)\ (0.60)}$$

$$= \frac{2}{13}.$$

EXERCISE 12.4

(1) A factory has two Machines-I and II. Machine-I produces 60% of items and Machine-II produces 40% of the items of the total output. Further 2% of the items produced by Machine-I are defective whereas 4% produced by Machine-II are defective. If an item is drawn at random what is the probability that it is defective?

(2) There are two identical urns containing respectively 6 black and 4 red balls, 2 black and 2 red balls. An urn is chosen at random and a ball is drawn from it. (i) find the probability that the ball is black (ii) if the ball is black, what is the probability that it is from the first urn?

(3) A firm manufactures PVC pipes in three plants viz, *X, Y* and *Z*. The daily production volumes from the three firms *X*, *Y* and *Z* are respectively 2000 units, 3000 units and 5000 units. It is known from the past experience that 3% of the output from plant *X*, 4% from plant *Y* and 2% from plant *Z* are defective. A pipe is selected at random from a day's total production,

(i) find the probability that the selected pipe is a defective one.

(ii) if the selected pipe is a defective, then what is the probability that it was produced by plant *Y* ?

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